Case 41: Custom fuel-parameter Rothermel sweep

1 Verification purpose

This case verifies ELMFIRE's public custom fuel-table interface and the Rothermel response to one-at-a-time changes in dead 1-h load, 1-h surface-area-to-volume ratio (SAV), fuel-bed depth, dead extinction moisture, and heat content. It can expose column misalignment, ignored custom values, stale precomputed coefficients, or unit conversion errors. The question is whether every generated code-250 row produces independently calculated reaction intensity and rate of spread (ROS) within 1% and 2%.

2 Mathematical formulation and numerical implementation

The custom row supplies class loads w_0, surface-area-to-volume ratios σ, depth δ, dead extinction M_x, and heat content h. These feed packing ratio $\beta = \sum w_0/(\delta\rho_p)$, optimum packing, reaction velocity, propagating flux, damping, heat sink, and

$$R_0 = I_R\xi/(\rho_b\epsilon Q_{ig}). \tag{1}$$

The formulation follows Andrews (2018), RMRS-GTR-371. Particle density and mineral constants are fixed by the inspected ELMFIRE interface and therefore are not claimed as user-sweepable variables.

3 Derivation of expected behavior

The baseline row is copied and exactly one public column is changed in each simulation condition. A case-specific fuel table is written beside that simulation condition and read independently by the reference calculation. The full equation is recalculated rather than imposing a monotonic trend: load, SAV, and depth have competing effects through packing and reaction velocity, so a simple ordering would be scientifically unjustified.

Each custom-fuel reference is recalculated from $R_0 = I_R\xi/(\rho_b\epsilon Q_{ig})$, changing only the indicated fuel-table column. The substitutions produced from the generated table are

Condition	R_{ref} (m min^{-1})	$I_{R,\text{ref}}$ (kW m^{-2})
baseline	0.411242	1045.499
$w_{1h} = 0.05$	0.253350	688.121
$w_{1h} = 0.15$	0.551674	1379.611
$SAV = 1000$	0.241692	960.426
$SAV = 3000$	0.662626	1074.561
$d = 0.5$	0.225975	910.468
$d = 2.0$	0.694647	991.942
$M_x = 15\%$	0.371790	945.198
$M_x = 35\%$	0.438500	1114.797
$h = 6000$	0.308432	784.125
$h = 10000$	0.514053	1306.874

For example, replacing the baseline 1-h load 0.10 by 0.05 lb ft^{-2} changes $R_0 = 1.349221$ to 0.831202 ft min^{-1}; multiplication by 0.3048 gives 0.253350 m min^{-1}. Each simulation condition uses $e_R = |R_{\text{max}} - R_{\text{ref}}|/\max(|R_{\text{ref}}|, 10^{-8})$ and the analogous e_I. The time of arrival (TOA) fit $t = a + bx$ gives $R_{\text{TOA}} = 60/b$ and $R^2 = 1 - SSE/SST$. The aggregate substitutions are $\max e_R \le 0.02$, $\max e_I \le 0.01$, $\max e_{\text{TOA}} \le 0.05$, and $\min R^2 \ge 0.995$.

4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for direct ROS and 0.01 for reaction intensity. Simulation conditions predicted at or above 0.10 m min^{-1} additionally require TOA-derived ROS error at most 0.05 and minimum $R^2 = 0.995$; slower simulation conditions are assessed from direct ROS and intensity. Pointwise equality to a complete recalculation detects both ignored columns and wrong nonlinear propagation. The baseline and ten low/high simulation conditions are all required.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE's supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive reference spread rate, the prescribed and maximum timesteps use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured Courant–Friedrichs–Lewy (CFL) limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE's stalled-front branch and corrupt the terminal timestamp. A zero-ROS simulation condition retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required simulation condition must reach the prescribed stop time in its generated namelist; a direct-only simulation condition may terminate early because lack of propagation is part of the state being tested. All simulation conditions use fuel code 250, 5/7/9% dead moisture, 90% live moisture, zero wind and slope, and 10 m cells. A finite ignition strip supplies a planar east face in the interior analysis corridor while remaining clear of the lateral grid boundaries. The baseline row uses 0.10/0.05/0.02 dead loads, 0.08 live-woody load, 1-h SAV 2000 ft^{-1}, depth 1 ft, dead extinction 25%, and heat content 8000 Btu lb^{-1}. Each output should match its own reference calculation; no universal increasing trend is predicted.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Custom fuel code	250	code	Selects simulation-specific table row
Changes	load, SAV, depth, extinction, heat	native table units	One-at-a-time interface exercise
Wind/slope	0 / 0	mph / degrees	Isolates custom fuel response
Timing	generated whole-second step and exact-multiple stop	s	Terminal-evidence compatibility

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

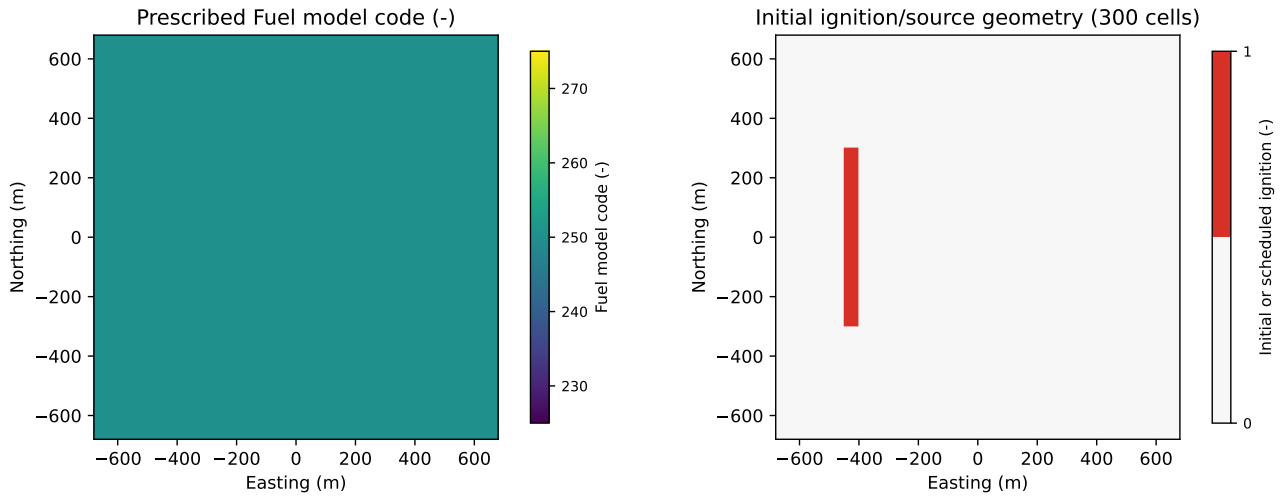


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The available results cover 11 of 11 simulation conditions and reports **PASS**.

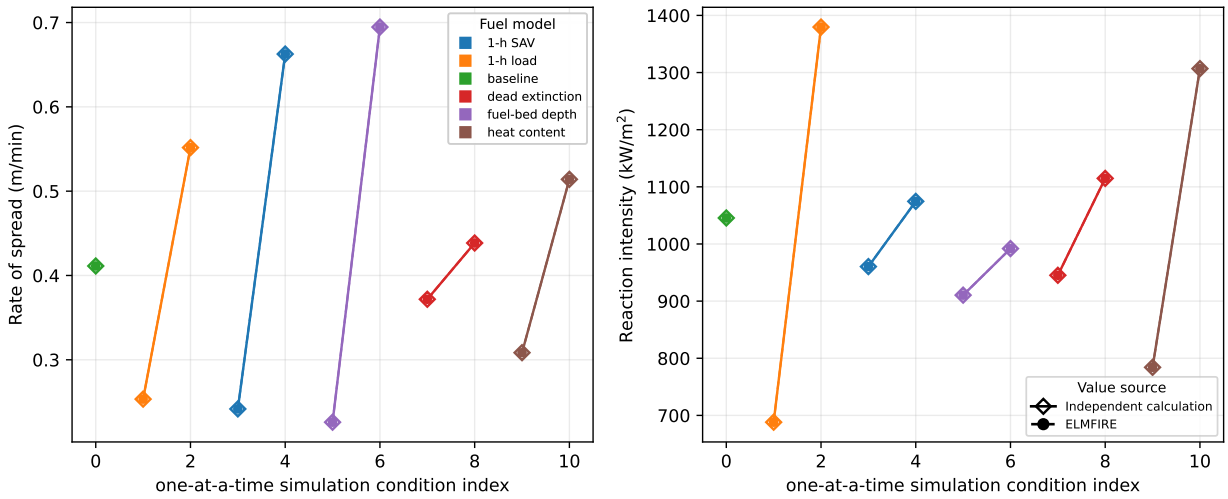


Figure 2: Independent calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines) for the custom fuel-table simulation conditions. Each comparison pair uses the same color.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	4.62672e-07 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	1.10118e-07 fraction	PASS
maximum TOA-ROS relative error	≤ 0.05	0.0151881 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999968 dimensionless	PASS

The authoritative decision is **PASS**.

8 Technical interpretation

If only one low/high pair fails, its fuel-table column or dependent coefficient is suspect. A common intensity error suggests table parsing or units; correct intensity with wrong ROS points to heat sink, density, or flux ratio. Identical results across a changed pair indicate the column was ignored.

9 Scope and limitations

This case tests only parameters exposed by the current custom table and one synthetic baseline. It does not test particle density, mineral contents, spatial mixtures, dynamic curing, or physical realism of the synthetic fuel.